Exact Small-Alphabet Distortion and Strict Alphabet Dependence for Length-Two Edit Distance

Ryutaro Yonezu

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Abstract

Let X_q be the space of all words over a q-letter alphabet of length at most two, including the empty word, with unit-cost Levenshtein edit distance, and let D_q be the least bi-Lipschitz distortion of X_q into an ℓ_1 space. We produce exact rational primal and dual cut-cone certificates establishing D_1 =1, D_2 =6/5, D_3 =17/14, and D_4 =123/101. A second computer-assisted integer cut inequality on the 31 words in X_5 proves $D_q \geq 785/643$ for every $q \geq 5$, and consequently $D_5 > D_4$. We give a universal explicit cut embedding with distortion at most 9/7, and prove $D_k \leq D_\infty \leq kD_k/(k-1)$ for $k \geq 2$ via random alphabet maps. The finite exhaustive certificates and source code are supplied for independent scrutiny. Priority relative to the finite-metric-embedding literature is not established.

1 Definitions

Let Σ_q ={0, ... ,q-1} and $X_q = \Sigma_q^{\leq 2}$. For u,v $\in X_q$, let d(u,v) be edit distance with insertion, deletion, and substitution each of unit cost. Define D_q as the infimum of t ≥ 1 such that a map f into a real ℓ_1 space satisfies

$$d(u,v) \leq \|f(u) - f(v)\|_1 \leq t\, d(u,v) \qquad (u,v \in X_q).$$

For a finite set X, each ℓ_1 metric is a nonnegative combination of cut semimetrics $\delta_C(u,v) = |1_C(u) - 1_C(v)|$. Hence D_q is a finite linear-programming optimum. We allow the empty word. There are $|X_q| = 1 + q + q^2$ words.

2 Main results

Theorem 1 (exact small-alphabet values).

$$D_1 = 1, \qquad D_2 = \frac{6}{5}, \qquad D_3 = \frac{17}{14}, \qquad D_4 = \frac{123}{101}.$$

Theorem 2 (strict alphabet dependence). For every integer q $\geq$ 5,

$$D_q \geq \frac{785}{643} > \frac{123}{101} = D_4.$$

In particular the optimum strictly increases between alphabets of sizes four and five. The inequality does *not* assert that 785/643 is the exact value of D_5 .

Theorem 3 (universal constructive upper bound). For every q $\geq$ 1,

$$D_q \leq \frac{9}{7}.$$

Theorem 4 (quantitative alphabet limit). Define $D_\infty := \sup_{q\geq 1} D_q(2)$, the supremum over *all finite alphabet sizes* with the word-length cap fixed at two. In this manuscript D_q is shorthand for D_q (2); D_∞ is **not** a limit in the word length. For every integer k $\geq$ 2,

$$D_k \leq D_\infty \leq \frac{k}{k-1} D_k.$$

In particular the limit exists and the convergence error satisfies

$$0 \leq D_\infty - D_k \leq \frac{9}{7(k-1)}.$$

3 Proof synopsis of Theorem 1

For q=2,3 enumerate all cuts up to complement, solve the cut-cone linear program, and provide an exact rational primal (embedding) certificate and nonnegative dual multipliers. Every cut inequality is checked using integer/fraction arithmetic by `code/verify_small.py`.

For q=4, the group generated by permutations of the four letters and word reversal has fifteen orbits of unordered word pairs. Averaging a cut over the group yields one column with coordinates b_i / N_i , where b_i is the number of separated pairs in orbit i and N_i is its size. Enumerating the 2^{20} cuts with the empty-word coordinate fixed to zero yields 13,882 distinct signatures, including zero. Nine cut-signature representatives give an exact rational primal certificate. Eight nonzero rational dual coefficients prove the matching lower bound. These independent checks are performed by `code/verify_exact.py` and `code/enumerate_cuts.cpp`.

4 Proof synopsis of Theorem 2

Let the fifteen q=4 pair orbits be indexed i=0, ... ,14, and let x_i be the average ℓ_1 distance over orbit i. The q=4 lower and upper LP dual multipliers are listed in `q4_exact_certificate.json`: the lower multipliers have weights 7/101 (i=4), 38/101 (i=7), 11/202 (i=9), and 29/202 (i=12); the upper multipliers have weights 15/101 (i=3), 45/101 (i=5), 32/101 (i=8), and 9/101 (i=11). The sum of upper weights times edit distance is 1; the sum of lower weights times edit distance is 123/101.

For each of the five four-letter subsets A of Σ_5 , apply this dual signed linear functional to the restriction of an ℓ_1 metric to X_A . Let S denote the sum of those five signed functionals. To see the bound used below, write the averaged distances on X_A as x_i . Noncontraction gives $x_i \geq d_i$, while distortion at most t gives $x_i \leq$ t d_i . Since all lower and upper dual multipliers are nonnegative, the signed functional satisfies

$$\begin{aligned} S_A &= \sum_i y_{\text{upper},i} x_i - \sum_i y_{\text{lower},i} x_i \\ &\leq t \sum_i y_{\text{upper},i} d_i - \sum_i y_{\text{lower},i} d_i = t - \frac{123}{101}. \end{aligned}$$

There are exactly five choices of A, hence $S = \sum_A S_A \leq 5(t - 123/101)$. This is an upper bound on a *signed functional* of the embedded metric, not on each cut individually.

Similarly define T on X_5 using the pair-orbit dual coefficients. The same nonnegative-multiplier argument gives $T \leq t - 47/37$ for every noncontracting distortion-t embedding: lower weights 10/37 (i=4), 27/74 (i=7), 5/37 (i=9), and upper weights 47/74 (i=5), 27/74 (i=11). Upper weights times edit distance sum to 1 and lower weights times edit distance sum to 47/37.

The key finite cut lemma is

$$606\, S(\delta_C) + 185\, T(\delta_C) \geq 0 \qquad \text{for every cut } C \text{ of } X_5.$$

The signed edge coefficients for S,T are *fully explicit* in `evidence/q5_ST_edges.txt`, using the common multiplier 896880. The verifier reconstructs all 465 coefficients and **explicitly checks that both S and T are invariant under every one of the 120 alphabet permutations and both choices of global word reversal**. This orbit invariance justifies the canonical 56-state reduction below. A cut and its complement give the same semimetric, so the empty-word bit is fixed to zero. Among the five (single-letter, doubled-letter) membership-bit pairs, relabeling of Σ_5 reduces 4^5 assignments to $\binom{8}{3} = 56$ multisets. For each, the 20 mixed-length-two words are freely assigned bits: hence $56 \cdot 2^{20} = 58{,}720{,}256$ cases are sufficient and exhaustive. `code/audit_q5_ST.cpp` evaluates the lemma with 64-bit signed integer arithmetic, periodically cross-checks its incremental Gray-code updates against direct evaluation, and records zero violations.

Since every ℓ_1 metric is a nonnegative cut combination, the lemma applies to any purported embedding metric. If this metric has distortion t, dual normalization gives $S \leq 5(t - 123/101)$ and $T \leq t - 47/37$. Consequently

$$0 \leq 606 \cdot 5 \left(t - \frac{123}{101}\right) + 185 \left(t - \frac{47}{37}\right),$$

so t $\geq 785/643$. X_5 is an isometric subspace of X_q for q ≥ 5. This proves the claim, conditional only on the exhaustiveness and correctness of the supplied finite integer verifier (which is open for audit).

5 Proof synopsis of Theorem 3

For each letter a, define $S_a = \{a\}$, $R_a = \{aa\}$, $L_a = \{ab : b \neq a\}$, and $T_a = \{ba : b \neq a\}$. Take cut semimetrics from the eight families

R	1/14
$S \cup L,\ S \cup T$	1/14 each
$R \cup L,\ R \cup T$	3/14 each
$S \cup R \cup L,\ S \cup R \cup T$	3/14 each
$S \cup R \cup L \cup T$	1/14

summed over letters a. Add global cuts `nonempty` with weight 9/14; `length exactly 2` with 3/14; and `repeated letters aa` with 2/14.

A pair of words uses at most four distinct letters. All these cut values depend only on equalities between the appearing letters; contributions from nonappearing letters vanish. Therefore the fifteen pair orbits for q=4 suffice to check the inequalities for all q ≥ 4, and q=1,2,3 are induced subspaces. Direct exact evaluation gives

$$d(u, v) \leq d_*(u, v) \leq \frac{9}{7} d(u, v).$$

The verifier computes the cut distance explicitly and checks all word pairs, including q=4, with Fraction arithmetic.

6 Proof synopsis of Theorem 4

Independently map every original letter uniformly to one of k colors. The induced map on words cannot increase edit distance. For each pair of words there are at most four distinct letters. Exhausting the fifteen equality patterns gives

$$\frac{\mathbb{E}[d(h(u), h(v))]}{d(u, v)} \in \left\{1, 1 - \frac{1}{k}, 1 - \frac{1}{2k}, 1 - \frac{1}{k} + \frac{1}{2k^2}\right\}.$$

Each expression is at least 1-1/k for k ≥ 2, and the bound is sharp for two distinct one-letter words. Compose this random map with a distortion- D_k embedding of X_k and take the resulting ℓ_1 direct-integral embedding. It has lower Lipschitz ratio at least 1-1/k and upper ratio at most D_k . Scaling by k/(k-1) proves Theorem 4. Explicit collision formulas are recorded in `evidence/collision_formulas.json`.

7 Prior work, audit scope, and limitations

7.1 Established finite-string computations

Andoni, Deza, Gupta, Indyk, and Raskhodnikova (2003) already studied unit-cost edit distance on finite binary strings and formulated optimal l1 distortion as a cut-semimetric linear program. In their computational-experiments section, they report distortion 4/3 for all binary strings of length at most three (a 15-point space, including the empty string). They also construct complete-bipartite obstructions yielding lower bounds approaching 3/2 as length grows. Thus finite-string distortion computations, cut decomposition, and the cut-LP method are established prior work. The current small cases and computational method are not claimed as new general techniques. Their reported length-three binary value concerns a different finite space from X_4 and X_5 at length cap two; it is not the assertion D_4 (2)=123/101 or the bound D_q (2) $\geq$ 785/643 for q $\geq$ 5. Failure to locate those constants in the sources inspected does not establish priority.

7.2 Relationship to OpenAI Math family 099

OpenAI's accompanying preprint and family-099 scope document address ordinary unit-cost insertion/deletion/substitution edit distance into real l1, with the maximum word length d growing. Their stated matching exponential-scale bounds hold for sufficiently large d, uniformly over finite alphabets. This manuscript instead fixes d=2 and varies alphabet size q, using exact rational finite certificates. It neither improves the asymptotic scale nor derives its length-two constants from the OpenAI theorem. Its proofs use the supplied finite cut certificates and elementary arguments; correctness of the OpenAI result is not a premise. No affiliation, endorsement, or independent verification of OpenAI's entire formalization is asserted.

7.3 Meaning of audit PASS

The independent AI audit supplied with the publication instruction reports PASS for the v0.2 mathematics and subsequent scope-preserving audit responses. The bundled evidence supports exact rational primal/dual checks, reconstruction of 465 integer edge coefficients, all 240 symmetry actions, 1,048,576 q4 cuts, and the 58,720,256-case canonical q5 scan. A separate Python/NumPy implementation reconstructs the q4 words, edit distances, orbits, signatures, and earliest cut representatives. These checks were rerun for this release. PASS is limited to these mathematical, code, and certificate checks; it is not a certification of novelty, priority, human peer review, or a fully formalized Lean/Coq proof.

The earlier supplied audit also reports a direct 2^{30} -cut q5 enumeration and a negative mutation test (185 changed to 186). The external implementation and run files are not bundled, and those two reported checks were not reproduced in preparing v1.0. They are not part of this package's independently reproducible evidence.

The lower bound is a computer-assisted proof: it relies on the correctness of the finite reduction, the supplied C++ integer checker, and the compiler/runtime executing it. Exact arithmetic avoids floating-point uncertainty in certificate verification, but is not a substitute for reviewing that logic. No human peer review has been performed. Novelty and priority remain INCONCLUSIVE.

7.4 Exact scope of the conclusions

The central conclusions are D_4 (2)=123/101 and D_q (2) $\geq$ 785/643 for every integer q $\geq$ 5, hence strict increase from four to five letters. The exact values of D_5 (2), D_∞(2) , and general D_q (2)

for $q \geq 5$ are not determined. The reported 457/367 upper bound and other stronger bounds from exploratory dialogue are not included as verified theorems because their complete frozen certificates have not been reconstructed here. All theorem statements and mathematical proof sections 1-6 are unchanged from v0.2. All verification code and frozen evidence files are byte-for-byte unchanged.

8 References

1. A. Andoni, M. Deza, A. Gupta, P. Indyk, and S. Raskhodnikova. Lower bounds for embedding edit distance into normed spaces. Proceedings of the Fourteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2003), pp. 523-526. DOI: 10.1145/644108.644196. Author-hosted full text: https://cs-people.bu.edu/sofya/pubs/embedding-of-edit-distance.pdf. See Sections 2-3 and Summary of computational experiments (PDF pp. 2-3).

2. OpenAI. Edit Distance in l1: Matching Bounds up to Constants in the Exponent. September 27, 2026. https://github.com/openai/math/tree/main/preprints/Edit-Distance-in-l1-Matching-Bounds-up-to-Constants-in-the-Exponent-September-27-2026.

3. OpenAI Math. Family 099: The sharp exponential scale of edit-distance distortion. Lean scope document. https://github.com/openai/math/blob/main/lean/docs/099.md. Accessed November 1, 2026. The scope document distinguishes selected binary lower-bound statements from companion upper-embedding results.

4. M. M. Deza and M. Laurent. Geometry of Cuts and Metrics. Algorithms and Combinatorics, vol. 15. Springer, 1997. DOI: 10.1007/978-3-642-04295-9. Background on cuts and metric cones; no novelty is claimed for the cut representation.

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